SAT Math: Hard-Module Challenge 1

Answer Key and Worked Solutions

Original, unofficial practice. Difficulty is not College Board-calibrated; this set does not predict an SAT score.

Answer Key

Question	Choice	Value	Question	Choice	Value
1	A	-4	7	C	5,256
2	C	52	8	B	1,216
3	B	19	9	B	$-26/5$
4	D	$11/31$	10	A	$26+10\sqrt{13}$
5	C	3	11	C	9
6	D	18	12	A	23

Worked Solutions

1. **A: -4.** For the system not to have a unique solution, its coefficient determinant must be zero:

$$(a-2)(a+1)-18=0 \quad\Longrightarrow\quad (a-5)(a+4)=0.$$

At $a=5$, the second equation is twice the first, so there are infinitely many solutions. At $a=-4$, the equations become

$$-6x+3y=7, \qquad 6x-3y=14.$$

Adding them gives $0=21$, so only $\boxed{a=-4}$ gives no solution.

2. **C: 52.** Since $f(2)=f(8)$, the axis of symmetry is $x=5$. Write $f(x)=a(x-5)^2+d$. Then

$$9a+d=24, \qquad a+d=-8,$$

so $a=4$ and $d=-12$. The requested sum is

$$a+b+c=f(1)=4(1-5)^2-12=\boxed{52},$$

not the constant term $f(0)=88$.

3. **B: 19.** Over the equal input intervals, the output differences are $35-11=24$ and $131-35=96$. Their ratio is

$$b^2=\frac{96}{24}=4,$$

so $b=2$. Now $a+c=11$ and $4a+c=35$ give $a=8$ and $c=3$. Thus $g(1)=8(2)+3=\boxed{19}$. Ratios of the original outputs do not give the growth factor because of the added constant.

4. **D: $\dfrac{11}{31}$.** Let p be the fraction of all components made by line B. Then

$$0.04(1-p)+0.12p=0.07 \quad\Longrightarrow\quad p=\frac{3}{8}.$$

The fraction of all components that are both from B and nondefective is

$$\frac{3}{8}(0.88)=\frac{33}{100}.$$

Since 93/100 of all components are nondefective, the requested probability is

$$\frac{33/100}{93/100}=\boxed{\frac{11}{31}}.$$

The denominator is the nondefective group, not the entire day's production.

5. **C: 3.** The original expression requires $x\neq 3$. Factoring reduces the equation to

$$\frac{(x-3)(x+3)}{x-3}=kx \quad\Longrightarrow\quad x+3=kx, \qquad x\neq 3.$$

At $k=1$, this gives $3=0$, so there is no solution. For $k\neq 1$, the candidate solution is

$$x=\frac{3}{k-1}.$$

This candidate is excluded when $k=2$, because it equals 3. Every other value of k gives one valid solution. The requested sum is $\boxed{1+2=3}$.

6. **D: 18.** At $k=3$, the equation is linear: $12x+3=0$, which has one solution. For $k\neq 3$, it is quadratic and has exactly one real solution when its discriminant is zero:

$$12^2-4(k-3)(3)=0 \quad\Longrightarrow\quad k=15.$$

Both values qualify, so their sum is $\boxed{3+15=18}$. The equation was not stated to remain quadratic for every k.

7. **C: 5,256.** Let n be the number of \$3 price increases. Ticket price is $30 + 3n$, and predicted sales are $240 - 12n$. After the \$6 per-ticket expense and the fixed expense, profit is

$$\begin{aligned} P(n) &= (30 + 3n)(240 - 12n) - 6(240 - 12n) - 1800 \\ &= (24 + 3n)(240 - 12n) - 1800 \\ &= -36n^2 + 432n + 3960 \\ &= 5256 - 36(n - 6)^2. \end{aligned}$$

Its maximum is $\boxed{\$5{,}256}$ at $n = 6$: a \$48 price and 168 tickets. Maximizing revenue alone would instead give a \$45 price and a smaller profit of \$5,220.

8. **B: 1,216.** The surface-area ratio $9 : 25$ gives a length ratio $3 : 5$ and a volume ratio

$$3^3 : 5^3 = 27 : 125.$$

Write the volumes as $27u$ and $125u$. Then

$$125u - 27u = 784 \quad \Longrightarrow \quad 98u = 784 \quad \Longrightarrow \quad u = 8.$$

Their sum is $(27+125)(8) = \boxed{1{,}216}$ cubic centimeters. The surface-area ratio cannot be applied directly to volume.

9. **B: $-\dfrac{26}{5}$.** Substitute $y = 2x + 1$ into the circle equation:

$$\begin{aligned} x^2 + (2x + 1)^2 - 10x + 6(2x + 1) + c &= 0, \\ 5x^2 + 6x + (7 + c) &= 0. \end{aligned}$$

Tangency means exactly one intersection, so the discriminant is zero:

$$36 - 20(7 + c) = 0 \quad \Longrightarrow \quad \boxed{c = -\frac{26}{5}}.$$

As a check, the circle has center $(5, -3)$ and radius squared $34 - c = 196/5 > 0$.

10. **A: $26 + 10\sqrt{13}$.** Set $AD = 4t$ and $DB = 9t$, with $t > 0$. The similar right triangles give

$$CD^2 = AD \cdot DB = 36t^2,$$

so $CD = 6t$ and $AB = 13t$. The area condition is

$$\frac{1}{2}(13t)(6t) = 156 \quad \Longrightarrow \quad t = 2.$$

Using the same triangle similarity, the legs are

$$AC = \sqrt{AD \cdot AB} = 4\sqrt{13}, \qquad BC = \sqrt{DB \cdot AB} = 6\sqrt{13},$$

while $AB = 26$. Their sum is $\boxed{26 + 10\sqrt{13}}$.

11. **C: 9.** Complete the square:

$$x^2 - 8x + 7 = (x - 4)^2 - 9.$$

For $k > 0$, the equation separates into

$$(x - 4)^2 = 9 + k \quad \text{or} \quad (x - 4)^2 = 9 - k.$$

The first always has two solutions. The second has two solutions for $0 < k < 9$, one for $k = 9$, and none for $k > 9$. The two branches cannot share a solution when $k > 0$. Thus exactly three distinct solutions occur only at $\boxed{k = 9}$.

12. **A: 23.** Square the given equation:

$$\left(2^x + 2^{-x}\right)^2 = 2^{2x} + 2(2^x)(2^{-x}) + 2^{-2x} = 25.$$

Since $(2^x)(2^{-x}) = 1$, the requested value is

$$2^{2x} + 2^{-2x} = 25 - 2 = \boxed{23}.$$

Omitting the cross term would incorrectly give 25.